

## Fit Time — SDSS

Parametric

Nonparametric

Foundation

| n    | Parametric       |                    |                  |                  |                  |                  |                     |                       |                  |                      |                        |                  | Nonparametric    |                  |                  |                  |                  |                  | Foundation       |                  |                  |                  |                  |
|------|------------------|--------------------|------------------|------------------|------------------|------------------|---------------------|-----------------------|------------------|----------------------|------------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|      | LinearGauss-Homo | LinearGauss-Hetero | Student-t        | LogNormal-Homo   | LogNormal-Hetero | Gamma-GLM        | LinGauss-Homo-Ridge | LinGauss-Hetero-Ridge | Student-t-Ridge  | LogNormal-Homo-Ridge | LogNormal-Hetero-Ridge | Gamma-GLM-Ridge  | FlexCode-RF      | BART-Homo        | BART-Hetero      | Flow-Spline      | Quantile-Tree    | CatMLP           | MDN              | TabPFN Native    | TabPFN 2.5       | RealTabPFN 2.5   | TabICL Quantiles |
| 500  | <0.01<br>(<0.01) | <0.01<br>(<0.01)   | <0.01<br>(<0.01) | 0.02s<br>(<0.01) | 0.02s<br>(<0.01) | 1.3s<br>(1.3s)   | 0.01s<br>(<0.01)    | 0.01s<br>(<0.01)      | 0.02s<br>(<0.01) | 0.02s<br>(<0.01)     | 0.03s<br>(<0.01)       | 0.29s<br>(0.27s) | 1.1s<br>(0.01s)  | 4.1s<br>(0.36s)  | 24.6s<br>(2.7s)  | 23.7s<br>(0.81s) | 9.1s<br>(5.8s)   | 2.0s<br>(0.09s)  | 8.9s<br>(4.9s)   | 1.1s<br>(0.63s)  | 0.72s<br>(0.33s) | 0.72s<br>(0.34s) | 4.4s<br>(2.8s)   |
| 1K   | <0.01<br>(<0.01) | <0.01<br>(<0.01)   | 0.01s<br>(<0.01) | 0.03s<br>(<0.01) | 0.03s<br>(<0.01) | 0.05s<br>(<0.01) | 0.01s<br>(<0.01)    | 0.01s<br>(<0.01)      | 0.02s<br>(<0.01) | 0.04s<br>(<0.01)     | 0.04s<br>(<0.01)       | 0.03s<br>(<0.01) | 1.6s<br>(0.05s)  | 5.3s<br>(0.47s)  | 30.0s<br>(2.8s)  | 39.4s<br>(1.9s)  | 6.3s<br>(2.1s)   | 3.1s<br>(0.05s)  | 5.9s<br>(0.29s)  | 0.72s<br>(0.26s) | 0.43s<br>(<0.01) | 0.42s<br>(<0.01) | 2.9s<br>(0.06s)  |
| 10K  | 0.02s<br>(<0.01) | 0.02s<br>(<0.01)   | 0.06s<br>(<0.01) | 0.27s<br>(<0.01) | 0.28s<br>(<0.01) | 0.23s<br>(<0.01) | 0.05s<br>(<0.01)    | 0.06s<br>(<0.01)      | 0.09s<br>(<0.01) | 0.30s<br>(<0.01)     | 0.30s<br>(<0.01)       | 0.20s<br>(<0.01) | 12.3s<br>(0.19s) | 32.2s<br>(2.1s)  | 2.3m<br>(12.5s)  | 4.1m<br>(11.1s)  | 32.6m<br>(32.2m) | 29.3s<br>(0.74s) | 30.3s<br>(1.0s)  | 3.2s<br>(0.37s)  | 2.7s<br>(0.02s)  | 2.8s<br>(0.03s)  | 40.3s<br>(1.6s)  |
| 50K  | 0.08s<br>(<0.01) | 0.09s<br>(<0.01)   | 0.28s<br>(<0.01) | 1.3s<br>(0.03s)  | 1.3s<br>(0.02s)  | 1.0s<br>(0.03s)  | 0.22s<br>(<0.01)    | 0.25s<br>(<0.01)      | 0.42s<br>(0.01s) | 1.4s<br>(0.03s)      | 1.4s<br>(0.02s)        | 0.94s<br>(0.02s) | 54.0s<br>(0.07s) | 3.0m<br>(2.3s)   | 13.0m<br>(11.8s) | 12.0m<br>(31.5s) | 2.7h<br>(1.4h)   | 3.2m<br>(24.1s)  | 1.8m<br>(7.9s)   | 36.4s<br>(0.64s) | 37.0s<br>(0.32s) | 37.2s<br>(0.64s) | —                |
| 100K | 0.15s<br>(<0.01) | 0.16s<br>(<0.01)   | 0.53s<br>(0.02s) | 2.4s<br>(0.02s)  | 2.5s<br>(0.01s)  | 1.9s<br>(<0.01)  | 0.45s<br>(0.03s)    | 0.48s<br>(<0.01)      | 0.79s<br>(0.02s) | 2.7s<br>(<0.01)      | 2.7s<br>(0.01s)        | 1.8s<br>(0.01s)  | 1.9m<br>(1.2s)   | 6.1m<br>(6.4s)   | 26.4m<br>(20.5s) | 17.7m<br>(5.6s)  | 2.1h<br>(59.0m)  | 10.1m<br>(1.5m)  | 3.4m<br>(8.6s)   | —                | —                | —                | —                |
| 250K | 0.38s<br>(<0.01) | 0.40s<br>(<0.01)   | 1.3s<br>(0.01s)  | 5.9s<br>(0.04s)  | 5.9s<br>(0.07s)  | 11.7s<br>(0.38s) | 1.3s<br>(0.01s)     | 1.5s<br>(<0.01)       | 2.3s<br>(<0.01)  | 6.9s<br>(0.05s)      | 7.1s<br>(0.04s)        | 6.4s<br>(0.03s)  | 5.7m<br>(13.9s)  | 17.2m<br>(2.4s)  | —                | 21.7m<br>(33.5s) | —                | 38.1m<br>(7.7m)  | 11.1m<br>(28.5s) | —                | —                | —                | —                |
| 500K | 0.89s<br>(0.01s) | 0.90s<br>(<0.01)   | 2.9s<br>(0.05s)  | 13.0s<br>(0.44s) | 13.0s<br>(0.45s) | 18.0s            | 3.7s                | 3.7s                  | 5.4s             | 16.0s                | 16.2s                  | 11.6s            | 14.7m            | 45.1m<br>(45.4s) | —                | 28.5m<br>(3.1m)  | —                | 1.5h             | 26.7m            | —                | —                | —                | —                |